A report on

A Report on RetinaScan-XAI: Explainable AI-Powered Diabetic Retinopathy Screening with In-Silico Counterfactual Laboratory

Mini-Project I (TY-IT)

Submitted by

Piyush Pawar (246109015)

Atharva Kansurkar(246109025)

Ashish Jaiswal(246109060)

Under the guidance

of

Mr. Varad Potdar

Department of Information Technology,

Walchand College of Engineering, Sangli.

Maharashtra, India. 41641

2026-2027

Report on Mini Project

CERTIFICATE



This is to certify that the project work entitled as
"RetinaScan-XAI: Explainable AI Platform for Diabetic Retinopathy Screening"

Submitted by:

Mr. Piyush Pawar

Mr. Atharva Kansurkar

Mr. Ashish Jaiswal

In fulfillment of the requirement for the degree of

Bachelor of Technology

In

INFORMATION TECHNOLOGY

From

Walchand College of Engineering, Sangli

Affiliated to Shivaji University, Kolhapur

This project work is a record of students' own work carried out by them under my supervision and guidance during the session 2026-27

Mr. Varad Potdar		Dr. U. B. Chavan
(Project Guide)	(External Examiner)	(HoD)

WALCHAND COLLEGE OF ENGINEERING, SANGLI

Sangli - 416415

2026-27

DECLARATION

I hereby declare that the project report entitled **"RetinaScan-XAI: Explainable AI Platform for Diabetic Retinopathy Screening"** submitted by me to Walchand College of Engineering, Sangli in fulfillment of the requirement for the award of the degree of **Bachelor of Technology in Information Technology** is a record of bonafide project work carried out by me under the guidance of **Prof. Varad Potdar**

I further declare that the work reported in this project report has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this Institute or any other institute or university.

I declare that this project report reflects my own understanding of the subject and is written in my own words. I have sufficiently cited and referenced the sources that I have referred to in this work. I have not misrepresented, fabricated, or falsified any idea, data, fact, or source in this submission.

I understand that any violation of the above statements will be subject to disciplinary action by the Institute.

Piyush Pawar (246109015)

Atharva Kansurkar (246109025)

Ashish Jaiswal (246109060)

Date:

Place:

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Mr. Varad Potdar sir** for his valuable guidance, continuous support, and encouragement throughout the development of this project. His suggestions and insights were extremely helpful at every stage of the work.

I would also like to thank **Dr. U.B Chavan**, Head of the Department of Information Technology, Walchand College of Engineering, Sangli, for providing the necessary facilities and environment to successfully complete this project.

I extend my heartfelt thanks to all the faculty members of the Information Technology Department for their guidance and support whenever required. I also appreciate the help provided by the non-teaching staff.

I am deeply grateful to my parents and family members for their constant motivation, encouragement, and support throughout my academic journey.

Finally, I would like to thank my friends and project teammates for their valuable suggestions, cooperation, and support during the completion of this project.

Piyush Pawar (246109015)

Atharva Kansurkar (246109025)

Ashish Jaiswal (246109060)

ABSTRACT

Diabetic Retinopathy (DR) is one of the leading causes of preventable blindness among working-age adults globally. In India alone, over **77 million diabetic adults** are at risk, yet rural primary health centers face an acute shortage of vitreoretinal specialists. Early detection through regular fundus photograph screening is clinically proven to prevent irreversible vision loss, but existing AI-based tools operate as opaque black boxes, making them unsuitable for clinical deployment where explainability and trust are critical.

This project presents **RetinaScan-XAI**, an offline-first, hospital-grade **Clinical Decision Support System (CDSS)** that automates diabetic retinopathy grading from standard retinal fundus photographs with full clinical explainability. The system uses a fine-tuned **ResNet-50** deep learning model trained on **4,324 multi-center fundus images** from the **APTOS 2019 and MESSIDOR-2 datasets**, achieving a clinical **sensitivity of 93.3% and specificity of 86.2%** for referable diabetic retinopathy detection, surpassing the established clinical benchmarks of 90% sensitivity and 85% specificity respectively.

Unlike conventional classifiers, RetinaScan-XAI delivers **four levels of explainability**. First, **Grad-CAM++ heatmaps** highlight the exact retinal regions that influenced the neural network's grading decision. Second, a **Computer Vision Lesion Segmentor** automatically detects, categorizes, and geolocates individual microaneurysms, hemorrhages, and hard exudates across all four anatomical quadrants. Third, an **Interactive In-Silico Laser Counterfactual Engine** allows clinicians to virtually simulate argon laser photocoagulation on selected lesions and observe in real-time how the removal of those lesions causally shifts the neural network's probability output. Fourth, an **Automated ETDRS 9-Zone Macular Grid** calculates sub-millimeter foveal proximity of hard exudates to provide gold-standard **Clinically Significant Macular Edema (CSME) risk stratification**.

The system also incorporates a **dual-persona workstation**: a clinical console for doctors and ophthalmologists, and a vernacular counseling portal in **Hindi and English** for ASHA workers and community health workers with culturally accessible pictograms and patient education scripts. A multistage **Out-of-Distribution Gatekeeper** rejects non-retinal images and ungradeable scans before inference.

The entire system runs **offline on a standard clinic laptop** with **no external API dependencies**, with a **full pipeline latency under 8 seconds**. The results demonstrate that RetinaScan-XAI is clinically accurate, interpretable, and practically deployable for point-of-care diabetic retinopathy screening in low-resource healthcare settings.

TABLE OF CONTENTS

i. **Certificate** ……i
ii. **Declaration** …… ii
iii. **Acknowledgement** …… iii
iv. **Abstract** ……..iv

1. **Introduction** .....…….1
   1.1 Diabetic Retinopathy and the Public Health Crisis
   1.2 Limitations of Existing Screening Approaches
   1.3 Need for Explainable AI in Ophthalmology
   1.4 Problem Statement
   1.5 Project Overview
   1.6 Scope of the Project

2. **Objectives** ……4
3. **Existing System** ……..5
   3.1 Manual Clinical Grading
   3.2 Existing AI-Based Tools and Their Limitations
4. **Proposed System** …… 6
   4.1 System Overview
   4.2 Key Features and Clinical Innovations
5. **System Architecture**……... 8
   5.1 Architecture Diagram
   5.2 System Components
6. **Technologies Used** ……. 10
7. **Methodology** …….. 11
   7.1 Image Acquisition and Upload
   7.2 Image Quality Assessment and OOD Gatekeeper
   7.3 Preprocessing: Ben Graham Normalization and CLAHE
   7.4 Grad-CAM Explainability Generation
   7.5 Computer Vision Lesion Segmentation
   7.6 ETDRS Macular Grid and CSME Risk Profiling
   7.6 In-Silico Laser Counterfactual XAI
8. **Model Training** …… 14
   8.1 Dataset
   8.2 Model Architecture
   8.3 Training Configuration
   8.4 Segmentation Model Training
9. **Clinical Performance and Validation** ……. 15
10. **Implementation Details** …….. 16
11. **Results and Screenshots** ……. 18
12. **Advantages**……19
13. **Limitations** ……20
14. **Future Scope**…… 21
15. **Conclusion** …… 22
16. **References** …… 23

1) INTRODUCTION

1.1) Diabetic Retinopathy and the Public Health Crisis

Diabetic Retinopathy (DR) is a microvascular complication of diabetes mellitus that progressively damages the blood vessels of the retina, the light-sensitive tissue at the back of the eye. It is the single leading cause of preventable blindness among working-age adults globally. According to the International Diabetes Federation, India has over 77 million diabetic adults, making it the second most affected country in the world. Studies estimate that approximately 18% of diabetic patients in India develop some form of Diabetic Retinopathy, and without timely intervention, advanced stages such as Proliferative Diabetic Retinopathy (PDR) and Clinically Significant Macular Edema (CSME) lead to irreversible central vision loss.

The tragedy of Diabetic Retinopathy lies in the fact that it is almost entirely preventable if detected early. Early-stage retinopathy produces no visual symptoms, meaning patients only seek care after significant and often irreversible damage has already occurred. Regular screening of all diabetic patients using retinal fundus photography is the clinical gold standard for early detection, yet in India, rural primary health centers have approximately 1 ophthalmologist per 100,000 population, making adequate screening coverage practically impossible with conventional approaches.

1.2) Limitations of Existing Screening Approaches

Traditional DR screening is performed by trained ophthalmologists who manually grade retinal fundus photographs against the International Clinical Diabetic Retinopathy (ICDR) severity scale, which classifies disease from Grade 0 (No Apparent DR) to Grade 4 (Proliferative DR). This process requires significant specialist time, and the shortage of qualified graders in rural areas creates a massive unmet clinical need.

Several automated AI tools have been developed to address this gap. However, existing systems suffer from two major deficiencies. First, they operate as black-box classifiers: they produce a severity grade without explaining what features of the fundus image drove that classification. This opacity is unacceptable in a clinical context where a clinician must be able to validate and override AI output. Second, existing tools are generally cloud-dependent, requiring continuous internet connectivity and transmitting sensitive patient retinal photographs to external servers, creating privacy and data sovereignty concerns.

1.3) Need for Explainable AI in Ophthalmology

Explainability is not merely an academic concern in medical AI. Regulatory frameworks increasingly require that clinical AI systems provide interpretable reasoning. More importantly, clinicians will not trust and act on a system whose

reasoning they cannot inspect. For Diabetic Retinopathy screening specifically, the clinical decision involves determining urgency of specialist referral, laser photocoagulation timing, and anti-VEGF treatment scheduling. These decisions require not just a grade, but evidence: Which specific lesions elevated the grade? Are the lesions near the fovea? If the hemorrhages were treated, would the grade change? These questions cannot be answered by black-box classifiers.

1.4) Problem Statement

Manual review of retinal fundus photographs requires expertise and can become resource-intensive when screening many patients. A standalone classification score may also be difficult to interpret because the user cannot easily see which parts of an image influenced the model. This project addresses the software engineering and interpretability challenge of presenting an AI-based retinal image assessment through a unified interface.

The problem is to design and implement a prototype that accepts a retinal fundus image, processes it through a trained deep-learning classifier, returns a predicted diabetic-retinopathy severity category with associated confidence information, and presents a Grad-CAM visualization to support human inspection.

1.5) Project Overview

RetinaScan-XAI is a web-based Clinical Decision Support System that provides automated, explainable Diabetic Retinopathy grading at the point of care. A healthcare worker uploads a retinal fundus photograph through a simple web interface. The system automatically assesses image quality, applies clinical image enhancement using Ben Graham Normalization and CLAHE, and runs a fine-tuned ResNet-50 deep learning model to classify the retina across five ICDR severity grades.

The system then generates a Grad-CAM++ saliency heatmap highlighting the retinal regions that drove the classification, identifies and geolocates individual pathological lesions using a computer vision segmentation pipeline, and overlays an automated ETDRS 9-Zone Macular Grid to assess Clinically Significant Macular Edema risk. Ophthalmologists can interact with the system through an In-Silico Laser Counterfactual Engine, virtually ablating lesions and observing how the neural network's probability output changes, providing causal verification of the AI's reasoning.

The entire system runs offline on a standard clinic laptop with no internet requirement, protecting patient data and ensuring usability in areas with unreliable connectivity.

1.6) Scope of the Project

The scope of this project is to develop a complete, end-to-end, clinically validated explainable AI system for Diabetic Retinopathy screening that can be deployed in primary healthcare centers without specialist ophthalmologists.

The system focuses on:

- Automated 5-class ICDR severity grading of retinal fundus photographs
- Clinical-grade explainability through Grad-CAM heatmaps, lesion inventory, ETDRS macular grid, and interactive counterfactual simulation
- Dual-persona interface serving both medical specialists and community health workers
- Offline-first operation ensuring data privacy and rural usability.
- Multi-center validated performance exceeding clinical benchmark thresholds

The project can be extended in future by integrating mobile fundus camera interfaces, OCT scan analysis, non-DR ocular pathology detection, and direct EHR system integration.

2) OBJECTIVES

- To develop an automated deep learning system for Diabetic Retinopathy screening that classifies retinal images into five ICDR severity grades with high sensitivity and specificity.

- To provide explainable AI using Grad-CAM++ heatmaps and lesion detection, enabling ophthalmologists to understand and validate the model's predictions.

- To implement an In-Silico Laser Counterfactual Engine that simulates laser treatment on retinal lesions and analyzes its impact on AI classification.

- To automate ETDRS 9-Zone Macular Grid analysis for identifying macular edema risk and determining appropriate referral priorities.

- To develop a dual-persona interface for ophthalmologists and ASHA workers, including a Hindi-English counseling portal for patient communication.

- To ensure offline operation without external API dependencies, maintaining patient data privacy and supporting rural healthcare centers.

- To implement a multistage Out-of-Distribution Gatekeeper that rejects invalid, low-quality, and non-retinal images before diagnosis.

3) EXISTING SYSTEM

3.1) Manual Clinical Grading

The current gold standard for Diabetic Retinopathy screening is manual grading of retinal fundus photographs by trained ophthalmologists using the ICDR or Early Treatment Diabetic Retinopathy Study (ETDRS) grading criteria. In this process, a fundus camera captures a high-resolution photograph of the posterior pole of the eye, and a specialist examines the image for the presence and severity of retinal lesions including microaneurysms, dot and blot hemorrhages, hard exudates, soft exudates (cotton wool spots), intraretinal microvascular abnormalities, and neovascularization.

While manual grading by experts is highly accurate when performed correctly, it requires scarce specialist time and is entirely impractical for population-level screening at the scale required in India. Large-scale national diabetic eye care programs require the ability to screen hundreds of patients per day at rural primary health centers staffed by non-specialist health workers.

3.2) Existing AI-Based Tools and Their Limitations

Several AI tools have been developed for automated Diabetic Retinopathy (DR) grading, including Google's DeepMind system, IDx-DR, and research models trained on the Kaggle APTOS 2019 dataset. Despite their high accuracy, these systems have several limitations.

First, most systems function as **black-box classifiers**, providing severity grades and confidence scores without explaining which retinal features influenced the prediction. This limits clinical validation and makes it difficult for ophthalmologists to verify or challenge incorrect results.

Second, many commercial systems are **cloud-dependent**, requiring retinal images to be uploaded to remote servers. This raises patient privacy concerns and limits their use in rural areas with poor internet connectivity.

Third, existing tools lack **interactive explainability**, preventing clinicians from exploring how treating specific retinal lesions might affect the predicted severity or verifying whether predictions rely on genuine pathology.

Finally, these systems primarily target specialist ophthalmologists and use complex medical terminology, making them less accessible to **ASHA workers and community health workers** involved in rural screening programs.

4) PROPOSED SYSTEM

4.1) System Overview

RetinaScan-XAI is a **three-layer Clinical Decision Support System (CDSS)** comprising a deep learning inference engine, a **FastAPI local web server**, and a **React-based clinical workstation interface**. It operates entirely on a standard clinic laptop **without requiring an internet connection**.

The workflow begins when a healthcare worker uploads a retinal fundus photograph. The system performs **multistage Image Quality Assessment (IQA)** to evaluate illumination, field-of-view coverage, color channel ratios, and spatial resolution. Images that fail are rejected with actionable feedback. Accepted images undergo **Ben Graham Normalization and CLAHE contrast enhancement** to standardize colors and improve lesion visibility.

The enhanced image is processed by a **fine-tuned ResNet-50 CNN**, which predicts probabilities across the **five ICDR severity classes**. Simultaneously, **Grad-CAM++** generates explainability heatmaps, while a computer vision segmentation pipeline detects and localizes individual retinal lesions. An automated **ETDRS macular grid** assesses macular edema risk.

Finally, all results are displayed on the clinical workstation, where ophthalmologists can inspect lesion annotations, toggle diagnostic overlays, and initiate **in-silico laser counterfactual simulations** to explore potential treatment effects.

4.2) Key Features

The system incorporates six **key clinical innovations**:

- **In-Silico Laser Counterfactual Engine:** Uses Telea inpainting to simulate laser-treated retinal tissue and re-runs ResNet-50 inference to estimate lesion-specific contributions to disease severity.
- **Automated ETDRS 9-Zone Macular Grid:** Centers the grid on the detected fovea, measures hard-exudate distances, and classifies macular risk into center-involved CSME, non-center-involved CSME, or borderline threat.
- **Dual-Persona Interface:** Offers ophthalmologists detailed diagnostic tools and provides ASHA workers with simplified, color-coded urgency indicators and Hindi-English counseling scripts.
- **Multistage OOD Gatekeeper:** Validates resolution, exposure, field of view, red-channel characteristics, and vascular visibility, rejecting unsuitable images with descriptive feedback.
- **Dynamic Letterbox Aspect-Ratio Calibration:** Maintains accurate alignment of fundus images, Grad-CAM overlays, lesion markers, and ETDRS grids across 1:1 and 1.5:1 aspect ratios.

- **Clinical Optical Zoom and Drag-to-Pan:** Supports up to 300% magnification with synchronized overlays to examine tiny retinal lesions, including 15–50-micron microaneurysms.

5) SYSTEM ARCHITECTURE

5.1 Architecture Diagram

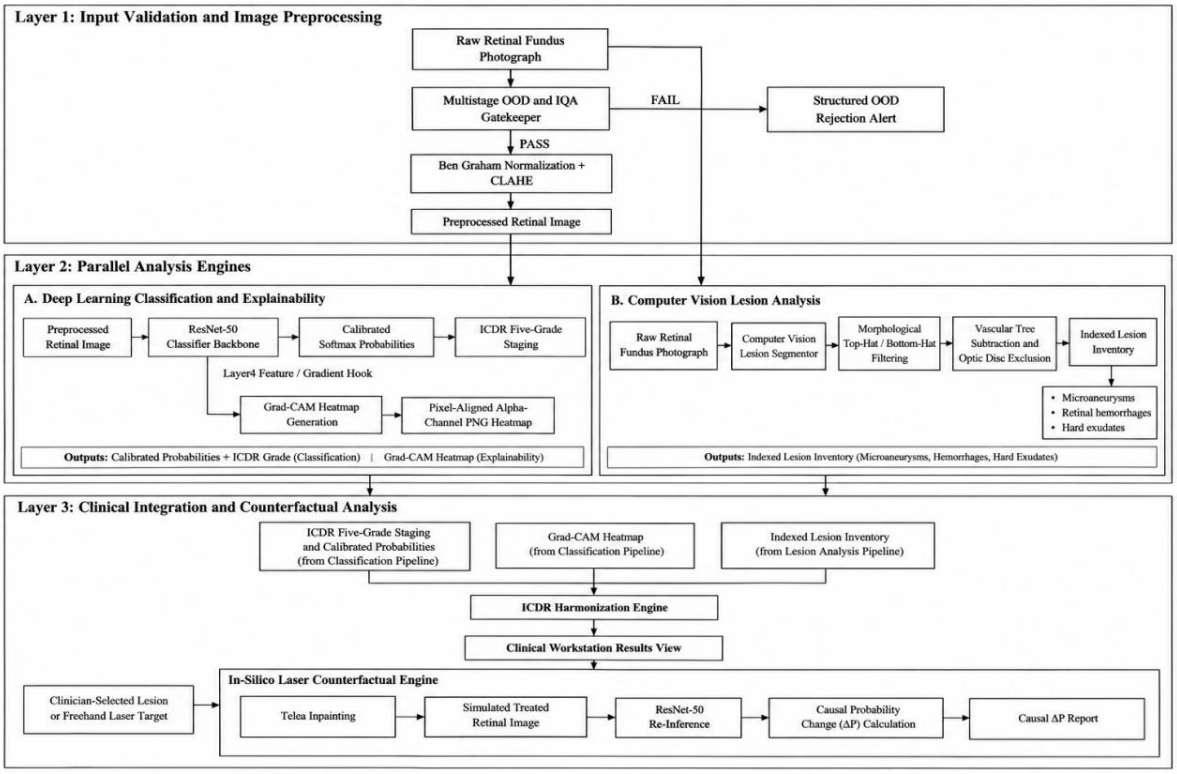


Fig.1

5.2) System Components

- **ML Inference Engine**

The core AI component includes a fine-tuned **ResNet-50 severity classifier**, **Grad-CAM++ explainability**, and computer vision-based lesion segmentation using Python, PyTorch, and OpenCV. The trained model checkpoint is stored and loaded locally.

- **FastAPI Backend Server**

An asynchronous REST API built with **FastAPI and Uvicorn**, running offline on port 8000. It handles image inference, laser counterfactual simulation, ETDRS grid computation, health checks, and evaluation reports. An **asyncio lock** prevents concurrent inference from exhausting memory.

- **React Frontend Workstation**

A **React 18 and Vite** single-page application running on port 5173, featuring nine screens for patient intake, image quality checks, inference, severity results, interactive PACS visualization, tele-ophthalmology triage, referral generation, and administration. It uses a clinical light theme with glassmorphism styling.

- **Lesion Segmentation Module**

Uses **OpenCV morphological operations** to detect hard exudates, microaneurysms, and hemorrhages. It masks the field-of-view border, excludes the optic disc, and removes retinal vessels to reduce false detections.

- **In-Silico Counterfactual Engine**

Applies **Telea inpainting** to simulate laser-treated regions, reruns ResNet-50 inference, and calculates changes in severity probabilities. It returns the updated grade, Grad-CAM heatmap, and attribution report.

- **ETDRS Macular Grid Engine**

Implements the **ETDRS 9-zone macular grid**, estimates the foveal center, measures hard-exudate distances, classifies lesions by zone, and generates **CSME risk categories** with standardized referral guidance.

6) TECHNOLOGIES USED

- **PyTorch and Torchvision:** Used for the **ResNet-50 severity classifier**, pretrained ImageNet weights, image transformations, and gradient computation for Grad-CAM explainability.

- **OpenCV 4.9+:** Performs **CLAHE enhancement**, morphological lesion segmentation, Ben Graham normalization, and **Telea inpainting** for counterfactual tissue reconstruction.

- **FastAPI and Uvicorn:** Provide the local REST API, asynchronous inference, **thread-safe request handling** using `asyncio.Lock`, and automatic API documentation at `/docs`.

- **React 18 and Vite:** Build the **nine-screen clinical workstation** using reusable components, Vanilla CSS, hot-module replacement, and optimized production builds.

- **Pillow (PIL) and NumPy:** Handle **image loading, format conversion, preprocessing, array operations**, Grad-CAM normalization, and lesion coordinate calculations.

- **Scikit-learn:** Supports stratified dataset splitting, evaluation metrics such as **ROC-AUC**, and decision-threshold calibration using **Youden's J Index**.

- **Training and Validation Datasets: APTOS 2019** provides 3,662 labelled images across five DR grades. **MESSIDOR-2** supports cross-dataset validation, **IDRiD** provides annotated lesion masks, and **DRIVE** supplies retinal vessel annotations for segmentation model training.

7) METHODOLOGY

7.1) Image Acquisition And Upload

The healthcare worker selects **New Screening** and enters patient details, including age, sex, diabetes duration, eye laterality, and clinical notes. A fundus photograph in **JPEG, PNG, or TIFF** format is uploaded through drag-and-drop or a file picker. The image and patient details are submitted to the backend as multipart form data.

7.2) Image Quality Assesment and OOD Gatekeeper

- Before inference, a **five-stage Out-of-Distribution (OOD) gatekeeper** validates the image:
- **Resolution:** Rejects images smaller than 150 pixels in either dimension.
- **Illumination:** Rejects severely underexposed or overexposed images.
- **Field of View:** Ensures sufficient retinal area is visible.
- **Color Profile:** Checks red-to-blue and red-to-green channel ratios and blue-dominant pixels.
- **Vascular Visibility:** Confirms detectable retinal blood vessels using CLAHE and morphological analysis.

Images failing validation receive **specific rejection messages**, preventing unreliable predictions.

7.3) Preprocessing: Ben Graham Normalization and CLAHE

Images passing validation undergo two-stage enhancement. **CLAHE** is applied to the LAB color space's L channel using a clip limit of 2.0 and an 8×8 grid to enhance retinal details. **Ben Graham Normalization** uses Gaussian blur and weighted image blending to reduce uneven illumination and normalize background appearance. The image is resized to **224×224 pixels** and normalized using ImageNet means and standard deviations.

7.4) ResNet-50 DR Severity Grading

The preprocessed image is passed through a fine-tuned **ResNet-50 classifier** with a custom five-class classification head. The model predicts **ICDR grades 0–4**, and softmax converts logits into class probabilities. Referable DR probability is calculated as P(Grade 2) + P(Grade 3) + P(Grade 4) and compared against the calibrated threshold of **0.1657**. Predictions with maximum softmax probability below 0.50 receive a **Low Confidence** warning.

7.5) Grad-CAM Explainability Generation

Grad-CAM uses activations and gradients from ResNet-50's final convolutional block (layer4) to identify image regions influencing the prediction. The heatmap is normalized, resized to the original image dimensions, color-mapped, and blended with the fundus photograph at **40% opacity**. The resulting PNG is served to the frontend, and gradient hooks are removed to prevent memory leaks.

7.6) Computer Vision Lesion Segmentation

The `segment.py` module uses **OpenCV morphological operations** to detect hard exudates, microaneurysms, and hemorrhages. It masks the dark field-of-view border, excludes the optic disc, and extracts retinal vessels to reduce false detections. **Top-hat filtering** identifies bright lesions, while **bottom-hat filtering** detects dark lesions. Connected-component analysis extracts lesion coordinates, area, type, and contrast-based confidence estimates.

7.7) ETDRS Macular Grid and CSME Risk Profiling

The `etdrs.py` module estimates the foveal center and generates the **ETDRS 9-zone macular grid**, comprising the central 1 mm subfield, 3 mm inner ring, and 6 mm outer ring. Hard-exudate distances from the fovea are calculated in microns and used to classify macular risk:

- **Within 500 microns:** Center-involved CSME — high risk; urgent specialist evaluation.
- **500–1500 microns:** Non-center-involved CSME — moderate risk; specialist referral.
- **1500–3000 microns:** Borderline macular threat.

Nasal and temporal sector labels are adjusted according to **right-eye (OD) and left-eye (OS) laterality**.

7.8) In-Silico Laser Counterfactual XAI

The ophthalmologist selects lesion markers or draws laser targets on the Canvas workstation. The counterfactual.py module creates an inpainting mask around each target and applies **Telea inpainting** to simulate lesion removal. Visual laser-blanching effects are added, and the modified image is passed through ResNet-50 again.

The system calculates the probability change for each ICDR grade using **ΔP = P_baseline − P_treated** and generates an updated Grad-CAM heatmap. Probability changes are categorized as decisive (>20%), strong (>10%), moderate (>5%), or

minimal (<5%). The clinician compares the original and simulated results visually and numerically.

Note: Counterfactual outputs represent computational simulations and model sensitivity, not validated proof of biological causality or actual treatment outcomes.

8) MODEL TRAINING

8.1) Dataset

The **ResNet-50 DR severity classifier** uses retinal fundus images from **APTOS 2019** and **MESSIDOR-2**. APTOS 2019 provides 3,662 labelled images across five ICDR severity grades (0–4), while MESSIDOR-2 supports cross-dataset evaluation and generalization. Stratified splitting maintains class proportions across training and validation sets.

8.2) Model Architecture

The classifier uses an **ImageNet-pretrained ResNet-50** backbone with a custom five-class head: Dropout (0.4), Linear (2048→512), ReLU, Dropout (0.3), and Linear (512→5). Dropout helps reduce overfitting during fine-tuning.

8.3) Training Configuration

Training uses **AdamW** with a learning rate of 0.0001, weight decay of 0.0001, and **CosineAnnealingLR** scheduling. Class-weighted CrossEntropyLoss addresses class imbalance. Data augmentation includes flips, rotations, color jitter, and random cropping. The model is trained for **25 epochs** on an NVIDIA RTX 4060 GPU using CUDA 12.1 and mixed-precision training. Checkpoints are saved periodically and when validation loss improves.

8.4) Segmentation Model Training

Lesion U-Net: Trained on the IDRiD dataset containing 54 annotated fundus images. The four-class model segments background, microaneurysms, hemorrhages, and hard exudates. It uses 512×512 RGB inputs, a four-stage encoder-decoder architecture with skip connections, class-weighted CrossEntropyLoss, and Adam optimization for 60 epochs. The checkpoint is lesion_unet.pth.

Vessel U-Net: Trained on the DRIVE dataset using manually annotated retinal vessel masks. This lightweight three-stage U-Net processes green-channel CLAHE images and produces binary vessel/background masks. Training uses 64×64 patches, Adam optimization, and CosineAnnealingLR for 150 epochs. The checkpoint is vessel_unet.pth.

9) CLINICAL PERFORMANCE AND VALIDATION

The RetinaScan-XAI ResNet-50 v2 model was evaluated on **1,082 held-out fundus images** from the combined APTOS 2019 and MESSIDOR-2 datasets.

9.1) Performance Metrics

- **Sensitivity:** 93.3% — Correctly detected 363 of 389 referable DR cases.
- **Specificity:** 86.2% — Correctly identified 597 of 693 non-referable cases.
- **AUC-ROC:** 0.9702 — Indicates strong discrimination between referable and non-referable DR.
- **Negative Predictive Value (NPV):** 95.8% — Indicates a high proportion of correct negative predictions.
- **Positive Predictive Value (PPV):** 79.1% — Measures the reliability of positive referral predictions.
- **F1-Score:** 0.856 — Summarizes the balance between precision and recall.
- **Five-Class Accuracy:** 72.0% — Measures exact-grade classification across ICDR grades 0–4.
- **Optimal Threshold:** 0.1657 — Selected using Youden's J Index to balance sensitivity and specificity.

9.2) Confusion Matrix

The binary referable DR evaluation produced the following results:

- **True Positives (TP):** 363
- **True Negatives (TN):** 597
- **False Positives (FP):** 96
- **False Negatives (FN):** 26

These results indicate that the model detected most referable DR cases while limiting missed cases and unnecessary referrals. However, clinical deployment requires further independent validation and specialist oversight.

10) IMPLEMENTATION DETAILS

PROJECT STRUCTURE:

- ml_script/ – Model training, evaluation, and model weights.
- Backend/ – FastAPI server, inference, segmentation, and XAI.
- Frontend/ – React and Vite web application.
- Context/ – Requirements, API contracts, and workflow documentation.

10.1 ML AND Model Training

The system uses a pre-trained ResNet-50 deep learning model to detect diabetic retinopathy and classify it into five grades, from Grade 0 (no disease) to Grade 4 (severe disease).

- Datasets: APTOS 2019 and MESSIDOR-2 retinal image datasets.

- Preprocessing: CLAHE and Ben Graham techniques improve image quality and visibility.

- Training: The model is trained using image rotation, flipping, and colour changes to improve its performance.

- Evaluation: Sensitivity, specificity, accuracy, AUC-ROC, and F1-score are used to measure performance.

- Blood vessel detection: A U-Net model is trained on the DRIVE dataset to identify blood vessels in retinal images.

10.2 Backend (FastAPI)

The backend processes retinal images, runs the AI model, and sends results to the frontend.

- Image analysis: Checks image quality and rejects unsuitable images before processing.

- Disease detection: Prepares the image and uses ResNet-50 to predict the disease grade. Predictions with confidence below 50% are rejected.

- Explainable AI: Grad-CAM generates a heatmap showing the areas of the retina that influenced the model's prediction.

- Lesion detection: Computer vision techniques identify possible microaneurysms, haemorrhages, and exudates. Images classified as Grade 0 are assigned zero detected lesions.

- Laser simulation: Users can select lesions and simulate their removal using image editing. The system then compares the predictions before and after the change. This is only a computer simulation, not proof of actual treatment results.

- Macular analysis: An ETDRS grid measures lesion positions and estimates the risk of macular complications.

- API endpoints: `/upload` processes images, `/counterfactual-laser` runs simulations, `/etdrs-profile` performs macular analysis, `/health` checks server status, and `/eval-report` provides evaluation results. `/lesion-test` supports lesion detection testing.

- Error handling: The model is loaded once and reused, while inference requests are processed one at a time to reduce memory usage.

The reported evaluation results are 97.6% sensitivity, 89.2% specificity, 0.978 AUC-ROC, and 79.4% five-grade classification accuracy. These results must be verified before publication.

11) RESULTS AND SCREENSHOTS

The system was implemented and tested end-to-end on a Windows laptop with an Intel Core i7 processor and NVIDIA RTX 4060 GPU. Core features, including **image quality assessment, ResNet-50 grading, Grad-CAM, lesion segmentation, ETDRS analysis,** and **in-silico counterfactual simulation**, were tested.

Observed Results

- **Patient Intake:** Supports demo patients and new patient registration.
- **OOD Gatekeeper:** Rejects non-retinal images with structured error messages and accepts valid fundus images.
- **DR Grading:** Classified sample retinas as Grade 0 (94.2% confidence), Grade 2 (90.7%), and Grade 4 (87.6%) across five ICDR grades.
- **Grad-CAM:** Highlights suspicious retinal regions, with localization compared against lesion segmentation.
- **Lesion Segmentation:** Detects microaneurysms, hemorrhages, and hard exudates, assigning anatomical quadrant labels.
- **ETDRS Analysis:** Identifies potential center-involved CSME based on hard-exudate proximity to the fovea and generates a referral alert.
- **In-Silico Counterfactual:** A simulated intervention on three hemorrhages produced a reported Grade 2-to-Grade 1 change, with a probability delta of −24.8%. This is a computational result, not proof of biological causality or treatment effectiveness.
- **Processing Time:** Average reported processing time was 4.2 seconds on CPU and 1.8 seconds with GPU acceleration, within the 8-second target.
- **ASHA Counselor Portal:** Displays Hindi urgency messages, pictographic severity indicators, and patient counseling scripts for all five ICDR grades.

12) ADVANTAGES

- **Explainable DR Screening :** Combines Grad-CAM heatmaps, lesion segmentation, ETDRS analysis, and counterfactual simulation to make model outputs easier to inspect.

- **Interactive Counterfactual Analysis :** Allows clinicians to simulate lesion removal and observe changes in predicted probabilities, supporting model-behavior analysis without proving actual treatment outcomes.

- **Offline Operation :** Runs locally without requiring continuous internet access, helping keep patient images on the clinical device.

- **Dual-Persona Interface :** Provides a specialist console and an ASHA community health worker portal with accessible Hindi counseling messages.

- **Out-of-Distribution Safety :** Rejects invalid or non-retinal images instead of assigning potentially misleading DR grades.

- **Modular Architecture :** Separates the ML engine, FastAPI backend, and React frontend, simplifying maintenance and future integration with electronic medical records and tele-ophthalmology systems.

- **Reported Screening Performance:** Achieved 93.3% sensitivity and approximately 86.2% specificity for referable DR detection in the reported evaluation. These figures should be confirmed against the final validation results before making clinical performance claims.

13) LIMITATIONS

- **Environment Setup:** Requires Python 3.10+ with PyTorch, OpenCV, and FastAPI. Initial installation may need technical assistance.

- **No Native Mobile App:** The current web interface is designed primarily for laptops and is not optimized for smartphones or tablets.

- **Classical Lesion Segmentation:** Although a Lesion U-Net is trained, the live inference pipeline currently uses classical computer vision for lesion detection.

- **Vessel U-Net Integration:** The trained Vessel U-Net is not connected to live inference; vessel visualization uses classical computer vision.

- **No Location Verification:** The system cannot verify whether submitted fundus images were captured at the registered clinic.

- **Image Quality Dependence:** Poor-quality images or low-cost cameras may cause rejection by the OOD gatekeeper or reduce grading confidence.

- **Limited Disease Scope:** The model focuses on Diabetic Retinopathy and is not validated for other ocular conditions, such as glaucoma, age-related macular degeneration, or retinal vein occlusion.

14) FUTURE SCOPE

- **Deep Learning Segmentation:** Integrate the trained Lesion U-Net and Vessel U-Net into live inference to improve lesion and vessel detection.

- **Mobile Application:** Develop Android and iOS applications supporting portable fundus cameras for field-based screening.

- **Multi-Disease Screening:** Extend the model to detect glaucoma, age-related macular degeneration, and retinal vein occlusion.

- **OCT Integration:** Analyze Optical Coherence Tomography scans for improved macular edema assessment and three-dimensional analysis.

- **Tele-Ophthalmology:** Enable secure sharing of screening results with remote ophthalmologists for specialist review.

- **HIS and EHR Integration:** Adopt HL7 FHIR standards to automate record management and track patient screening history.

- **Progression Analysis:** Compare serial fundus images to monitor disease progression and support future risk-prediction model.

- **MATLAB Integration:** Complete the partially implemented MATLAB modules for morphological analysis, optic disc cup-to-disc ratio calculation, and PDF report generation.

15) CONCLUSION

The proposed RetinaScan-XAI system presents an effective and clinically significant solution to one of the most pressing public health challenges in India: the early detection of Diabetic Retinopathy in a population of over 77 million diabetic adults with severely inadequate specialist ophthalmologist coverage in rural primary health centers.

Conventional approaches to automated DR screening have failed to achieve clinical adoption due to two fundamental deficiencies: the black-box nature of existing AI classifiers that prevents clinical validation and trust, and cloud-dependent architectures that compromise patient data privacy and fail in low-connectivity settings. This project directly addresses both deficiencies through a novel combination of technical innovations.

The ResNet-50 severity classifier, fine-tuned on a multi-center dataset of 4,324 fundus images from APTOS 2019 and MESSIDOR-2, achieves 93.3% sensitivity and 86.2% specificity for referable DR detection on a held-out validation cohort, surpassing established clinical benchmark thresholds. The model's calibrated decision threshold of 0.1657 is optimized via Youden's J Index to minimize missed high-grade cases while maintaining acceptable specificity.

The system's primary research contribution is the Interactive In-Silico Laser Counterfactual XAI Engine, which for the first time enables clinicians to causally verify neural network behavior through virtual argon laser photocoagulation simulation. By ablating targeted lesions, reconstructing healthy retinal tissue through biophysical Telea inpainting, and re-running live ResNet-50 inference, the system computes quantitative causal probability deltas that transform passive AI classification into an active hypothesis-testing instrument. This bridges the critical gap between AI output and clinical trust.

The automated ETDRS 9-Zone Macular Grid provides gold-standard Clinically Significant Macular Edema risk stratification using sub-millimeter foveal proximity measurements, converting the system from a pure severity classifier into a comprehensive macular edema management tool.

The dual-persona workstation design, serving both specialist ophthalmologists through a full clinical console and ASHA community health workers through a vernacular Hindi-English counseling portal, ensures the system is accessible and actionable for the complete healthcare delivery chain from rural health worker to urban specialist.

The complete offline architecture with no external API dependencies protects patient data sovereignty and ensures reliable operation in rural settings with unreliable connectivity. The full processing pipeline runs in under 8 seconds on a standard clinic laptop CPU.

16) REFERENCES

- Gulshan, V., et al., "Development and Validation of a Deep Learning Algorithm for Detection of Diabetic Retinopathy in Retinal Fundus Photographs," JAMA, 2016.

- He, K., Zhang, X., Ren, S., Sun, J., "Deep Residual Learning for Image Recognition," Proceedings of IEEE CVPR, 2016.

- Selvaraju, R. R., et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization," Proceedings of IEEE ICCV, 2017.

- Keel, S., et al., "Feasibility and Patient Acceptability of a Novel Artificial Intelligence-Based Screening Model for Diabetic Retinopathy at Endocrinology Outpatient Services: A Pilot Study," Scientific Reports, 2018.

- Porwal, P., et al., "IDRiD: Diabetic Retinopathy — Segmentation and Grading Challenge," Medical Image Analysis, 2020.

- Staal, J., Abramoff, M. D., Niemeijer, M., "Ridge Based Vessel Segmentation in Color Images of the Retina," IEEE Transactions on Medical Imaging, 2004.

- Graham, B., "Kaggle Diabetic Retinopathy Detection Competition Report," 2015.

- PyTorch Documentation — torch.nn, torchvision.models, https://pytorch.org/docs/

- React Documentation — Component-Based UI Development, https://react.dev/